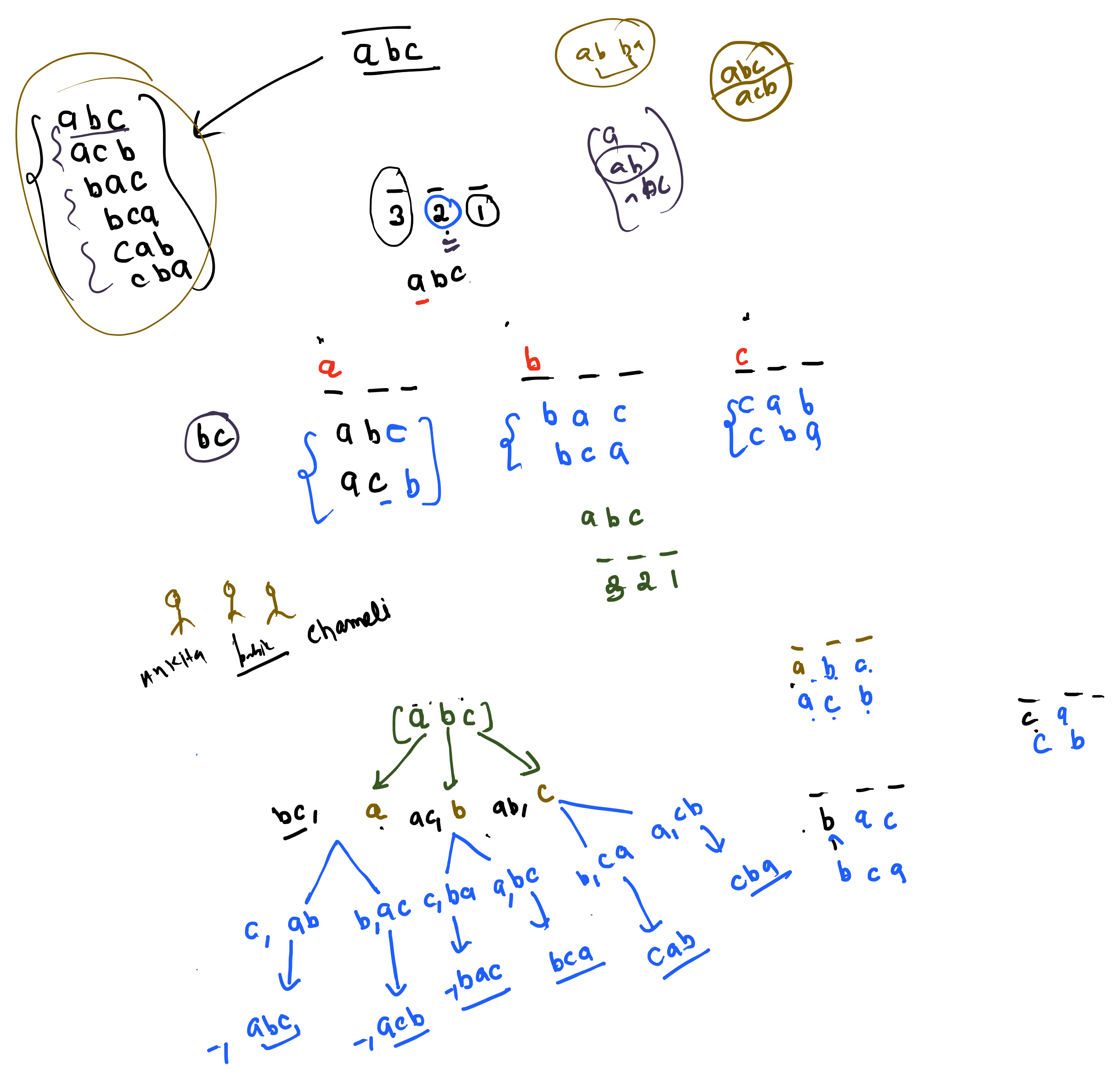
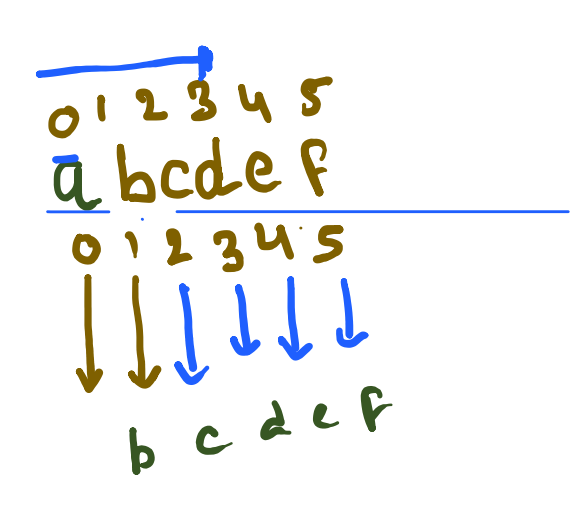
Lec-21

23/06/26 7:08 PM

Permutation



abc
abc
acb
bac
bca
cab
cba
3 2 1
abc
ab ba
abc acb
a
ab
abc
a _ _
bc
abc
acb
b _ _
bac
bca
c _ _
cab
cba
abc
3 2 1
nikita babli chameli
[abc]
bc, a
ac, b
ab, c
c, ab
b, ac
c, ba
a, bc
b, ca
a, cb
-1 abc
-1 acb
-1 bac
bca
cba
cab
a b c
a c b
b a c
b c a
c a
c b



0 1 2 3 4 5
a b c d e f
0 1 2 3 4 5
b c d e f

ans + charAt(i)

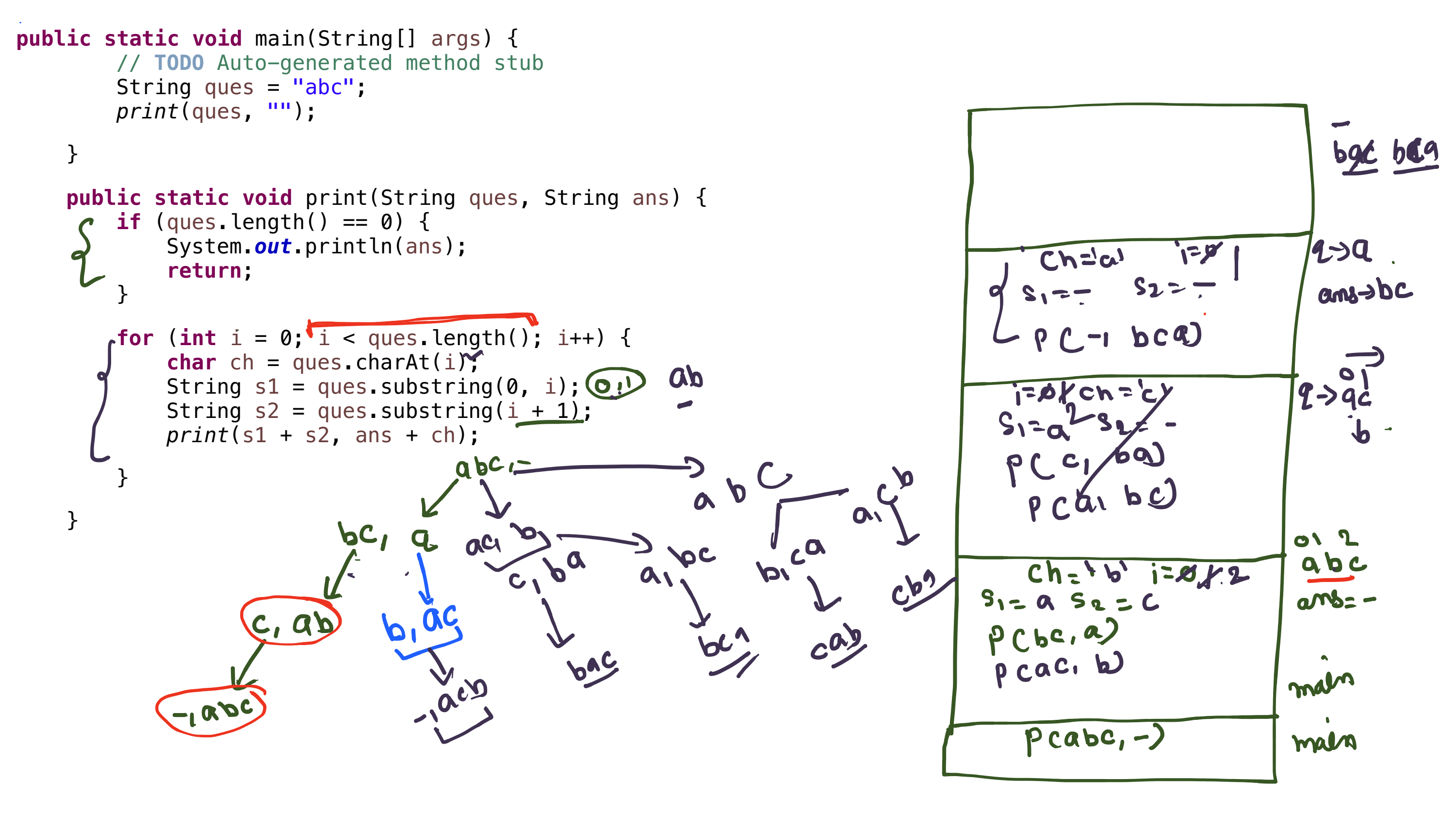
bcdef

```
for(i=0 ; i<ques.length() ; i++){
    ch = charAt(i)
    s1 = (0,i)
    s2 = (i+1)
    P(s1+s2, ans+ch)
}
```

s1 = (0,i)
s2 = (i+1)

Question

i=0	S1 = - (0,0)	S2 = bcdef (1) S1+S2 = bcdef	ans+'a'
i=1	S1 = a [0,1]	S2 = cdef [2] S1+S2 = acdef	ans+'b'
i=2	S1 = ab [0,2]	S2 = def [3] S1+S2 = abdef	ans+'c'
i=3	S1 = abc (0,3)	S2 = ef (4) S1+S2 = abcef	ans+'d'
i=4	S1 = abcd (0,4)	S2 = f (5) S1+S2 = abcdf	ans+'e'
i=5	S1 = abcde (0,5)	S2 = - (6) S1+S2 = abcde	ans+F



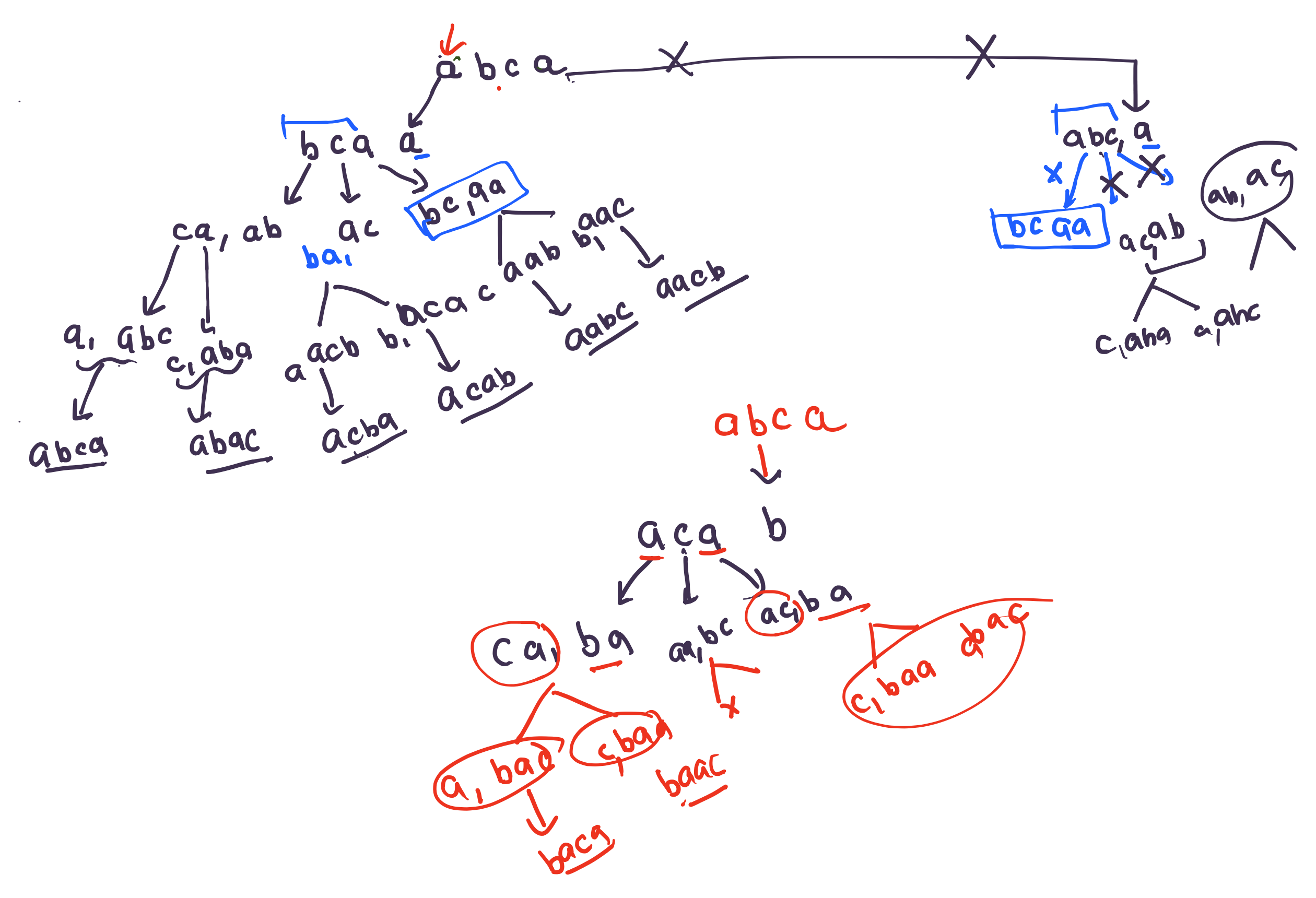
```
public static void main(String[] args) {
		// TODO Auto-generated method stub
		String ques = "abc";
		print(ques, "");

	}

	public static void print(String ques, String ans) {
		if (ques.length() == 0) {
			System.out.println(ans);
			return;
		}
		for (int i = 0; i < ques.length(); i++) {
			char ch = ques.charAt(i);
			String s1 = ques.substring(0, i);
			String s2 = ques.substring(i + 1);
			print(s1 + s2, ans + ch);
		}
	}
```

0,1 ab
abc, -
bc, a
c, ab
-1 abc
b, ac
-1 acb
ac, b
c, ba
bac
a b c
a, bc
b, ca
a, cb
bca
cab
cba
Ch='a' i=0 1
S1=- S2=-
P(-, bca)
i=0 1 ch='c'
S1=a S2=-
P(c, ba)
P(a, bc)
Ch='b' i=0 1 2
S1=a S2=c
P(bc, a)
P(ac, b)
P(abc, -)
bc bca
a→a
ans→bc
a→ac
b
0 1 2
abc
ans=-
main
main

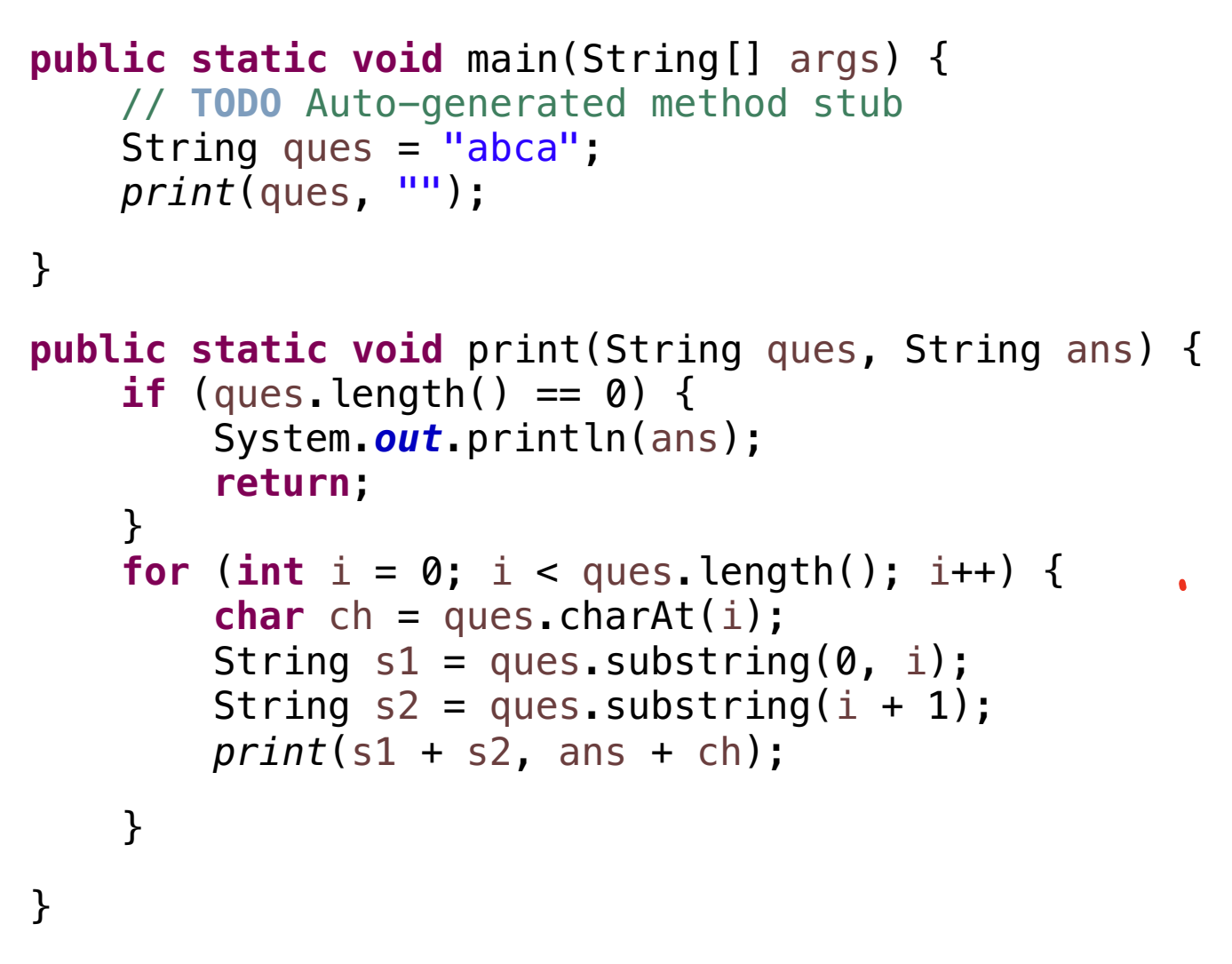
```
abca
abac
acba
acab
aabc
aacb
baca
baac
bcaa
bcaa
baac
baca
caba
caab
cbaa
cbaa
caab
caba
aabc
aacb
abac
abca
acab
acba
```

abca

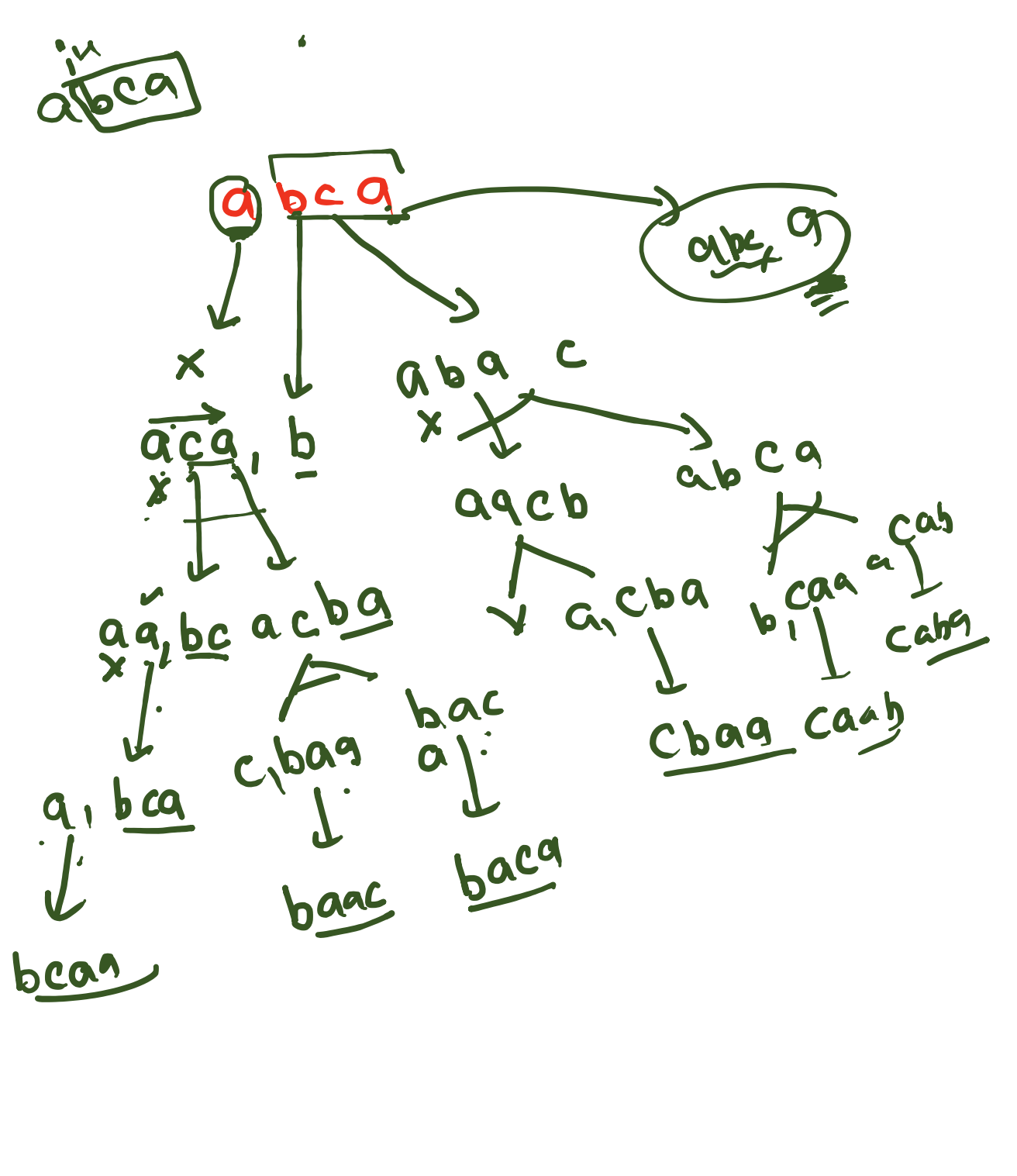
4! / 2!



abca
bca, a
ca, ab
a, abc
abca
c, aba
abac
ac
ba,
acb
acba
bc, aa
b, aca
acab
aab
aabc
aac
b,
aacb
abc, a
bc aa
a, ab
c, aba
a, abc
ab, aa
abca
aca b
ca, ba
a, bac
c, baa
baca
baac
a, bc
ac, ba
c, baa
bac



```
public static void main(String[] args) {
		// TODO Auto-generated method stub
		String ques = "abca";
		print(ques, "");

	}

	public static void print(String ques, String ans) {
		if (ques.length() == 0) {
			System.out.println(ans);
			return;
		}
		for (int i = 0; i < ques.length(); i++) {
			char ch = ques.charAt(i);
			String s1 = ques.substring(0, i);
			String s2 = ques.substring(i + 1);
			print(s1 + s2, ans + ch);
		}
	}
```

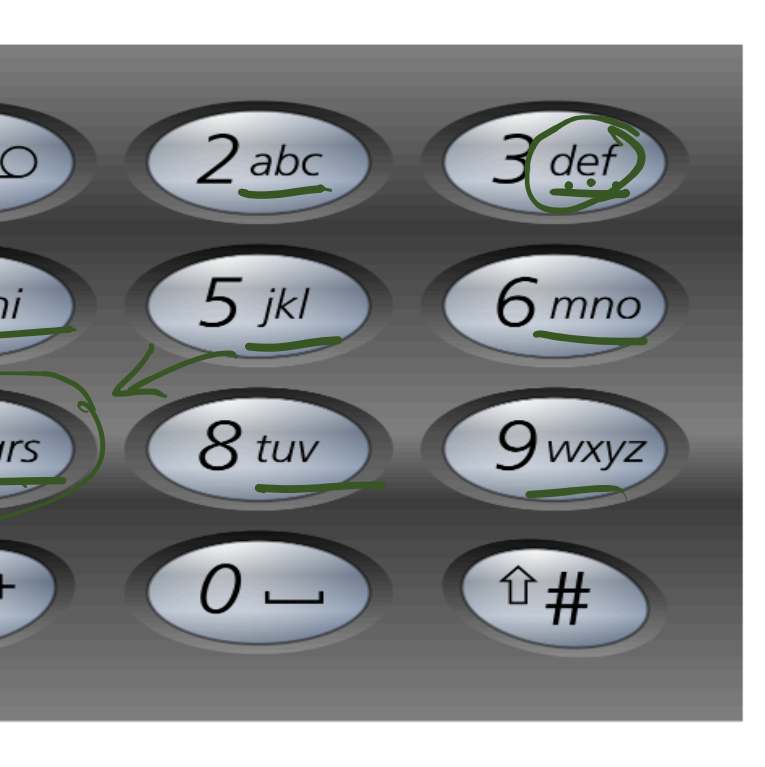


abca
abca
abc, a
aca, b
aba c
abc a
aa, bc
acba
aacb
a, cba
b, caa
cas
cabs
a, bca
bcaa
c, baa
baac
bac
a
baca
cbaa
caab

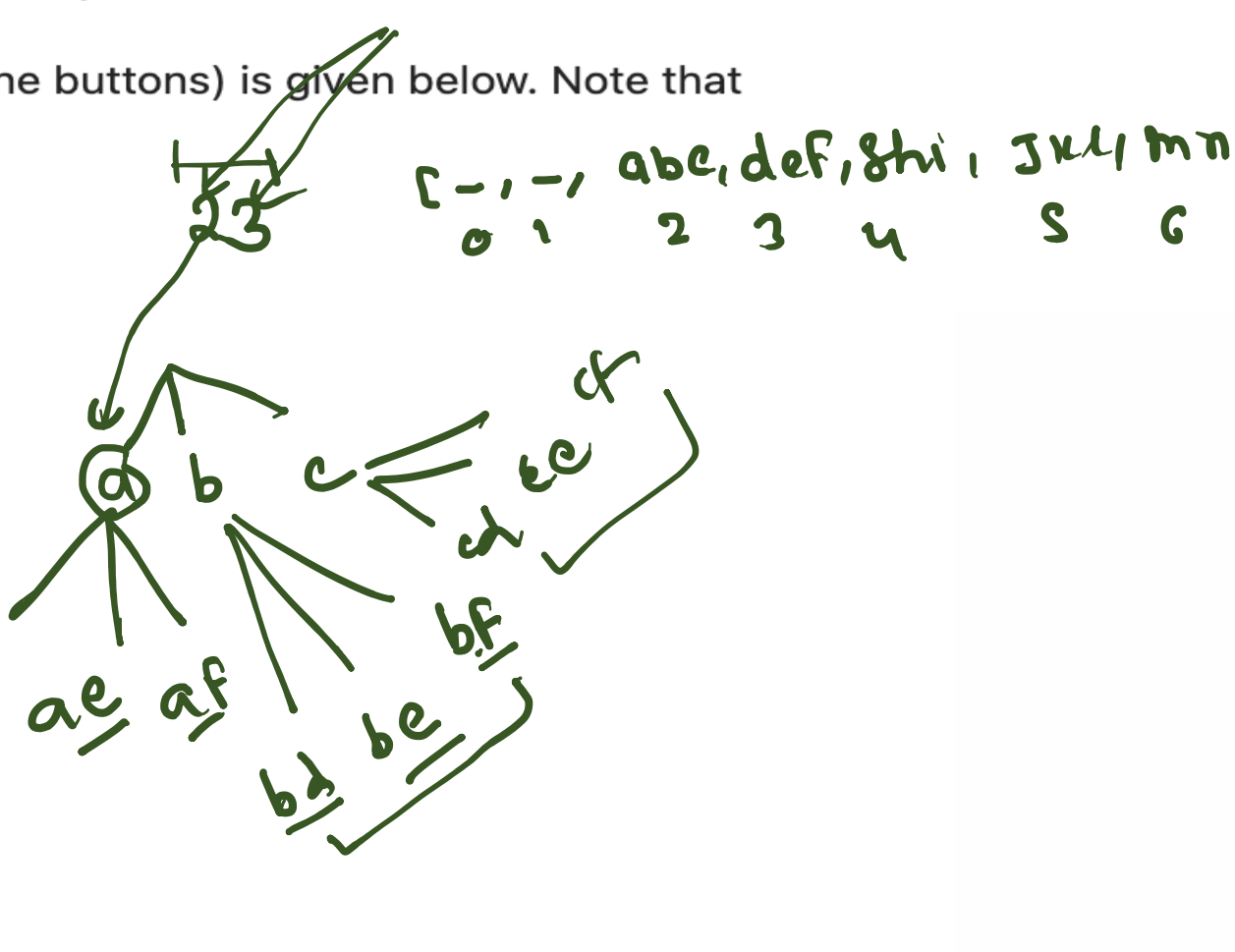
Given a string containing digits from `2-9` inclusive, return all possible letter combinations that the number could represent. Return the answer in **any order**.

A mapping of digits to letters (just like on the telephone buttons) is given below. Note that 1 does not map to any letters.

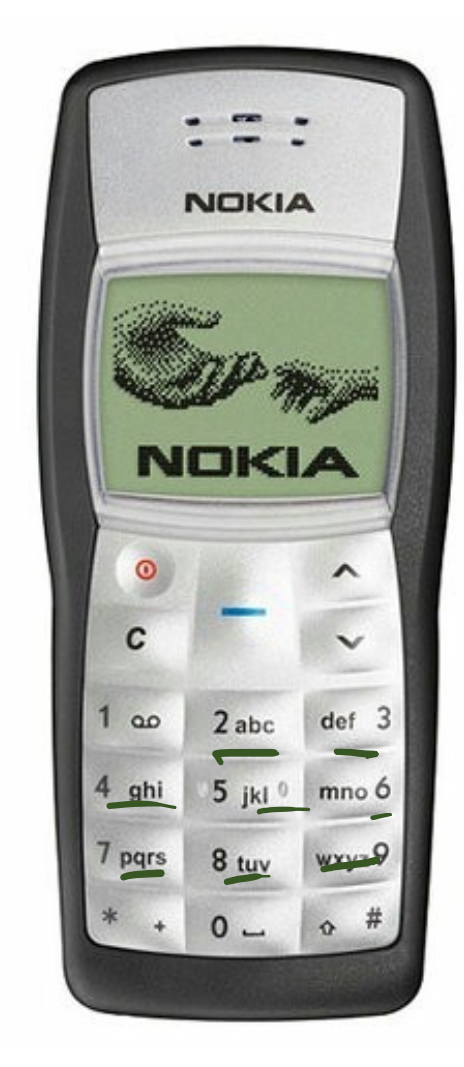
[-, -, abc, def, ghi, jkl, mno, pqrs, tuv, wxyz]
0 1 2 3 4 5 6 7 8 9



1
2 abc
3 def
4 ghi
5 jkl
6 mno
7 pqrs
8 tuv
9 wxyz
*
0
#



23
a
b
c
ad
ae
af
bd
be
bf
cd
ce
cf

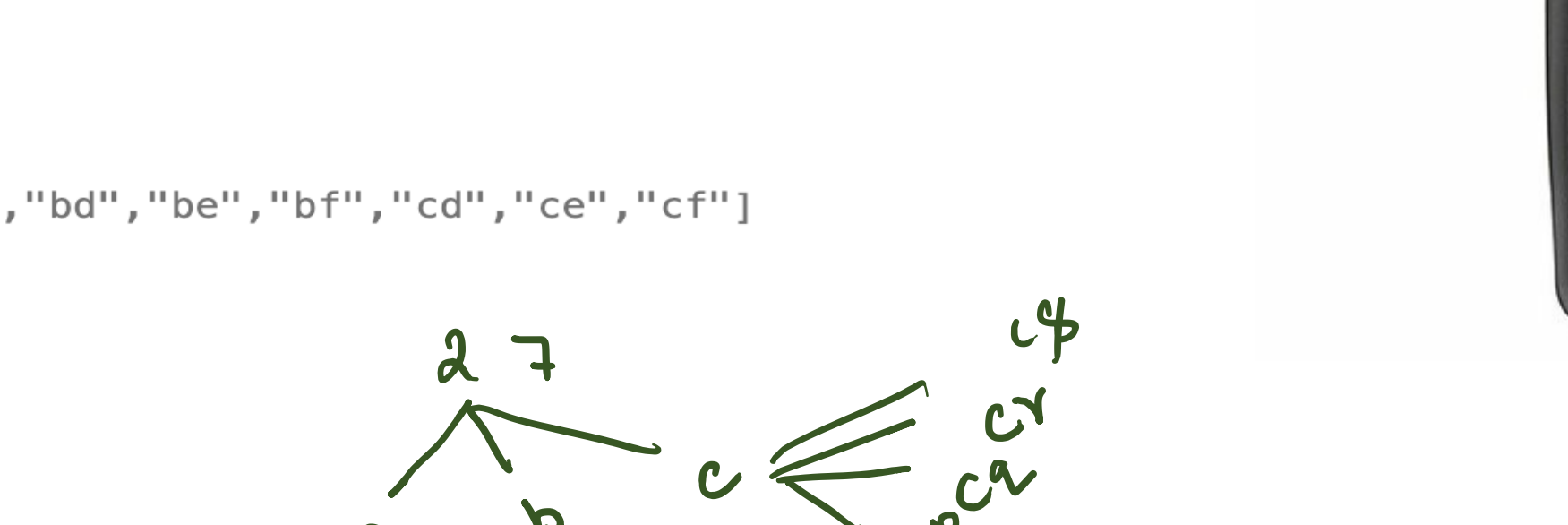


NOKIA
NOKIA

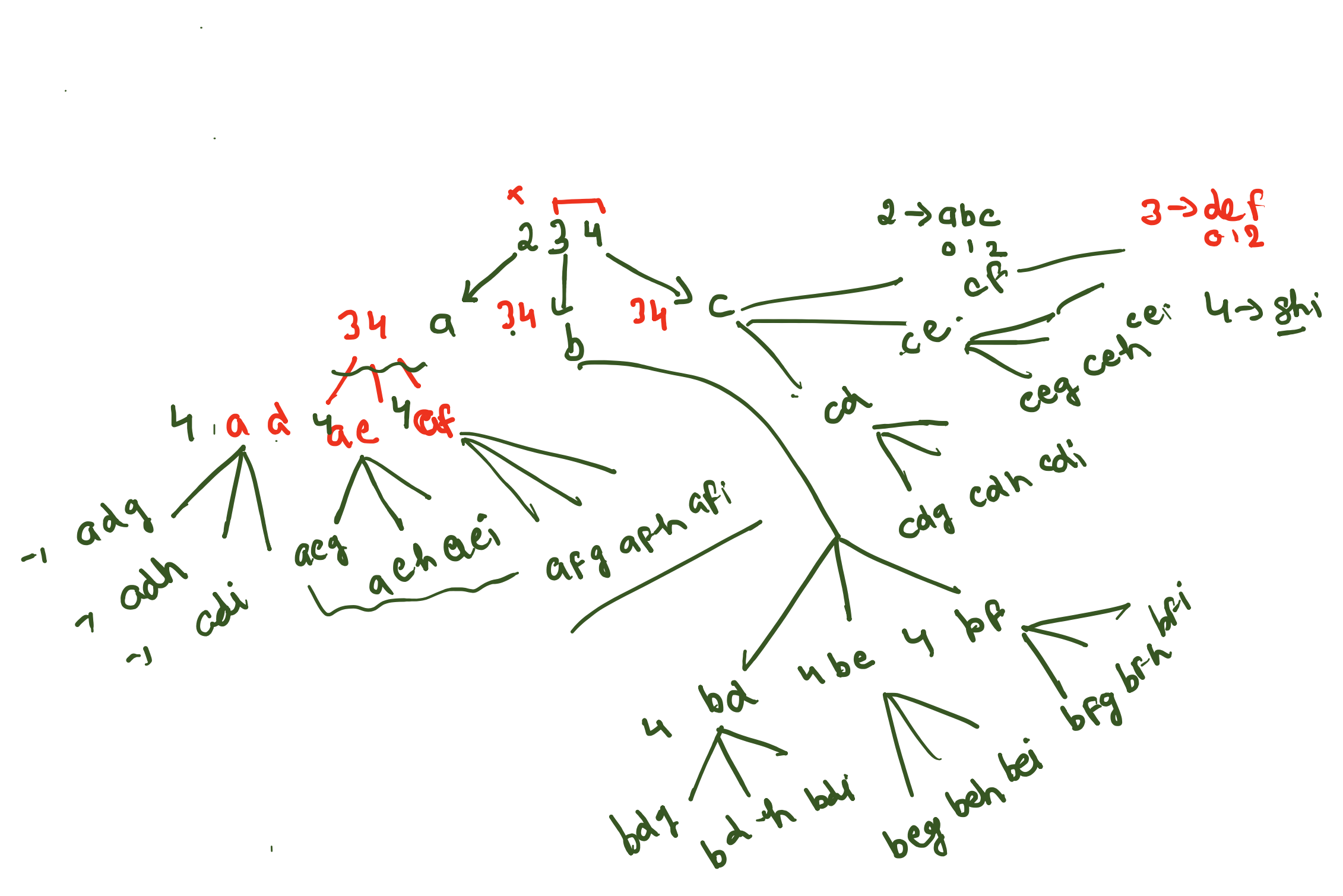
Example 1:

> Input: digits = "23"
> Output: ["ad","ae","af","bd","be","bf","cd","ce","cf"]

Example 2:



27
a
b
c
ap
aq
ar
as
bp
bq
br
bs
cp
cq
cr
cs

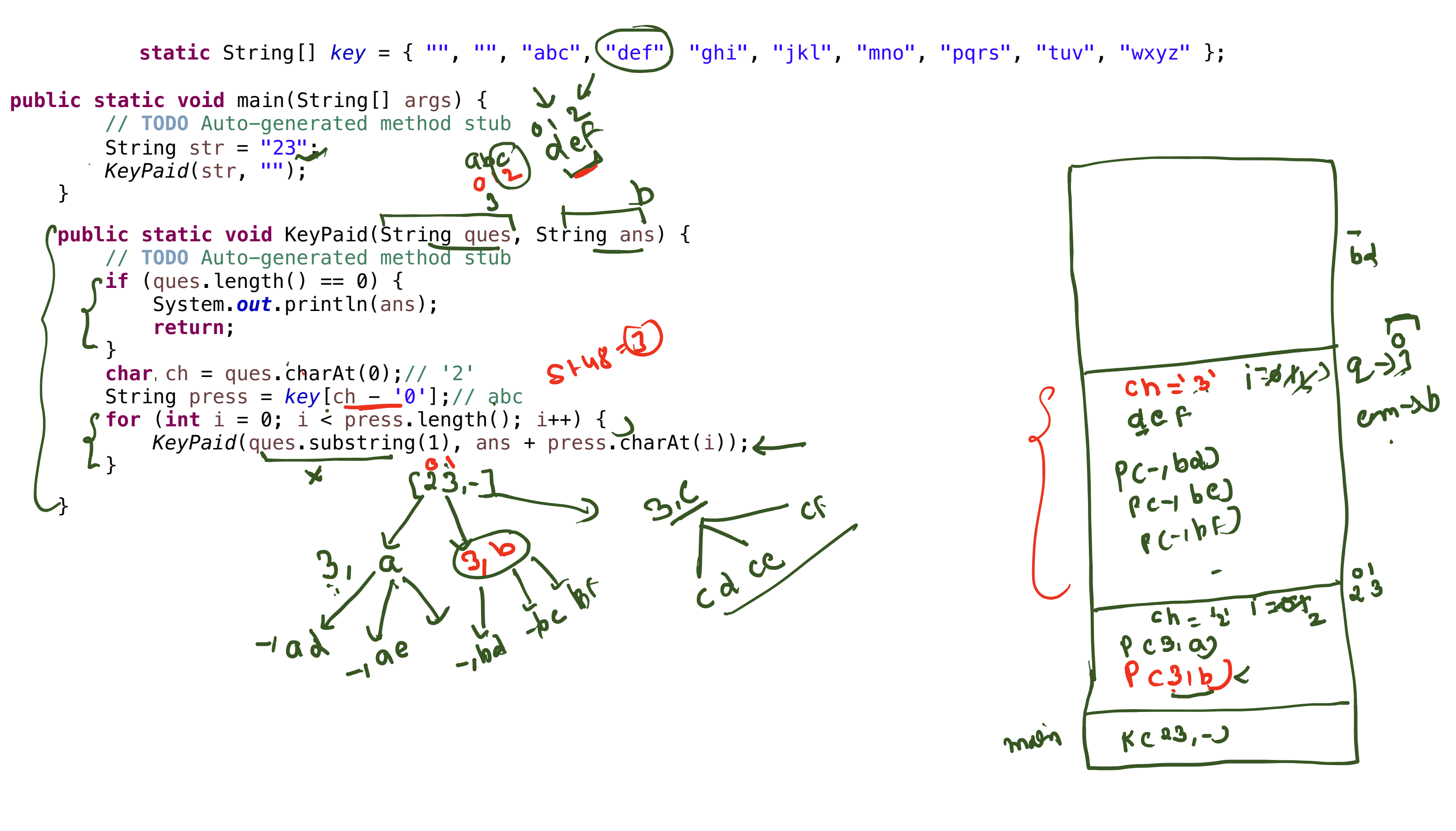


234
34 a
34 b
34 c
4 ad
4 ae
4 af
-1 adg
-1 adh
-1 adi
aeg
aeh
aei
afg
afh
afi
cd
ce
cf
ceg
ceh
cei
cdg
cdh
cdi
4 bd
4 be
4 bf
bdg
bdh
bdi
beg
beh
bei
bfg
bfh
bfi
2→abc
0 1 2
3→def
0 1 2
4→ghi

0 1 2
234
34 a
34 b
34 c
press

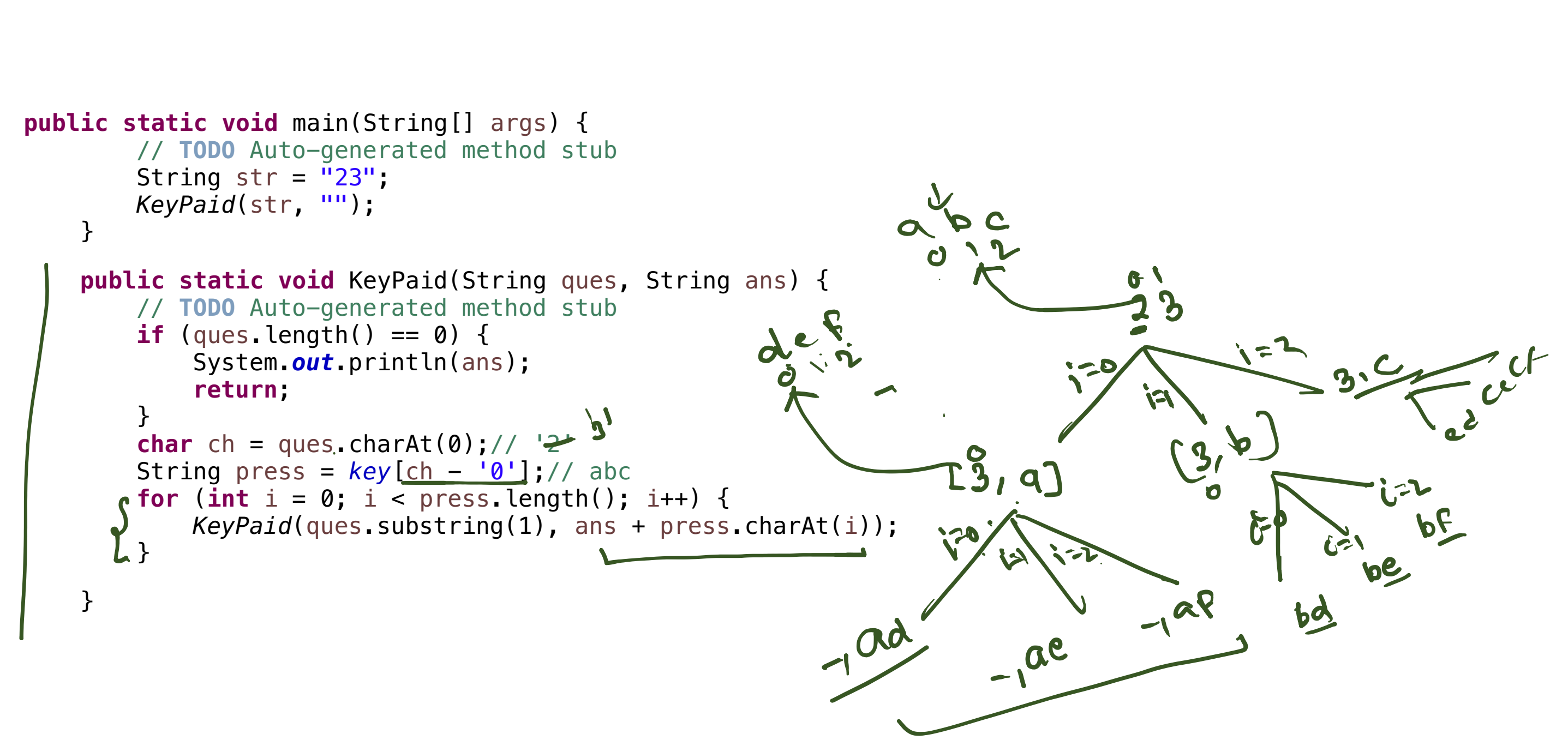
ch = q.charAt(0)
'2' → abc
0 1 2

```
For(i=0    i < press.len()    i++){
    F(q.sub(1), ans+press.charAt(i))
}
```



```
	static String[] key = { "", "", "abc", "def", "ghi", "jkl", "mno", "pqrs", "tuv", "wxyz" };

	public static void main(String[] args) {
		// TODO Auto-generated method stub
		String str = "23";
		KeyPaid(str, "");
	}

	public static void KeyPaid(String ques, String ans) {
		// TODO Auto-generated method stub
		if (ques.length() == 0) {
			System.out.println(ans);
			return;
		}
		char ch = ques.charAt(0);// '2'
		String press = key[ch - '0'];// abc
		for (int i = 0; i < press.length(); i++) {
			KeyPaid(ques.substring(1), ans + press.charAt(i));
		}

	}
```

abc
0 1 2
def
0 1 2
3
Step 1
[23, -]
3, a
3, b
3, c
-1 ad
-1 ae
af
-1 bd
-1 be
bf
cd
ce
cf
ch='3' i=0 1 2
def
P(-, bd)
P(-, be)
P(-, bf)
ch='2' i=0 1 2
P(3, a)
P(3, b)
main
K(23, -)
bd
q→3
ans→b
0 1
23



```
	public static void main(String[] args) {
		// TODO Auto-generated method stub
		String str = "23";
		KeyPaid(str, "");
	}

	public static void KeyPaid(String ques, String ans) {
		// TODO Auto-generated method stub
		if (ques.length() == 0) {
			System.out.println(ans);
			return;
		}
		char ch = ques.charAt(0);// '2'
		String press = key[ch - '0'];// abc
		for (int i = 0; i < press.length(); i++) {
			KeyPaid(ques.substring(1), ans + press.charAt(i));
		}

	}
```

abc
0 1 2
def
0 1 2
23
i=0
i=1
i=2
[3, a]
[3, b]
3, c
i=0
i=1
i=2
-1 ad
-1 ae
-1 af
bd
be
bf
cd
ce
cf